STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 8: Mechanical Properties of Solids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define stress. [1]
2. SI unit of Young's modulus? [1]
3. Longitudinal strain? [1]
4. Hooke's law? [1]
5. Modulus for volume change? [1]
6. Is strain dimensionless? [1]
7. Meaning of large Young's modulus? [1]
8. Longitudinal stress? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. 100 N on 2 x 10^-4 m^2: stress? [2]
10. 2 m wire extends 1 mm: strain? [2]
11. Stress=2 x 10^7 Pa, strain=10^-4: Y? [2]
12. 10 cm cube compresses by .1 cm: strain magnitude? [2]
13. L=1 m,A=2 x 10^-6 m^2,Y=2 x 10^11,F=40 N: extension? [2]
14. Why use steel reinforcement in concrete? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. Wire L=2m,A=10^-6,Y=2 x 10^11,F=100N: extension? [3]
16. Stress 4 x 10^7 Pa at strain 2 x 10^-4. Estimate modulus. [3]
17. B=1.5 x 10^9 Pa, volume strain=-2 x 10^-4. Pressure increase? [3]
18. Why does doubling wire length double its extension if F,A,Y fixed? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Distinguish proportional and elastic limits. [4]
20. Derive extension and calculate for F=50 N,L=1.5m,A=10^-6,Y=2 x 10^11. [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations and relations

- Stress sigma = F/A; longitudinal strain epsilon = Delta L/L (dimensionless).
- Young’s modulus Y = (F/A)/(Delta L/L) = FL/(A Delta L), so Delta L = FL/(AY).
- Bulk modulus B = -Delta p/(Delta V/V); therefore Delta p = -B(Delta V/V).
- Shear modulus G = shear stress/shear strain; shear strain approximately x/L for small deformation.
- Hooke’s law: stress proportional to strain within the proportional limit.
- For a cube, volume strain = Delta V/V; for small uniform linear compression, use the stated volume change directly.
- Stress and elastic moduli are in Pa (N/m^2); strain has no unit.

Use equations only when their assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Concept questions are answered directly; numerical problems show the governing equation and substitution.

Question 1

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Restoring force per unit area.
Final answer: Restoring force per unit area.

Question 2

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Pa.
Final answer: Pa.

Question 3

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: DeltaL/L.
Final answer: DeltaL/L.

Question 4

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Within proportional limit, stress is proportional to strain.
Final answer: Within proportional limit, stress is proportional to strain.

Question 5

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Bulk modulus.
Final answer: Bulk modulus.

Question 6

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Yes.
Final answer: Yes.

Question 7

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Material is relatively stiff.
Final answer: Material is relatively stiff.

Question 8

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: F/A.
Final answer: F/A.

Question 9

Working: Use sigma=F/A. Substitute F=100 N and A=2x10^-4 m^2: sigma=100/(2x10^-4)=5x10^5 Pa.
Final answer: 5 x 10^5 Pa.

Question 10

Working: Strain=Delta L/L. Convert 1 mm=0.001 m; strain=0.001/2=5x10^-4 (dimensionless).
Final answer: 5 x 10^-4.

Question 11

Working: Young’s modulus Y=stress/strain=(2x10^7)/(10^-4)=2x10^11 Pa.
Final answer: 2 x 10^11 Pa.

Question 12

Working: Volume strain magnitude=|Delta V|/V=0.1 cm/10 cm=0.01. It is dimensionless.
Final answer: 0.01.

Question 13

Working: Delta L=FL/(AY)=40(1)/[(2x10^-6)(2x10^11)]=40/(4x10^5)=1x10^-4 m.
Final answer: 1 x 10^-4 m.

Question 14

Working: Concrete is strong under compression but relatively weak under tension. Steel reinforcement carries tensile stress and helps the composite structure resist bending and cracking.
Final answer: Steel carries tensile stress effectively; concrete is strong in compression.

Question 15

Working: Delta L=FL/(AY)=100(2)/[(10^-6)(2x10^11)]=200/(2x10^5)=1x10^-3 m=0.001 m.

Final answer: 0.001 m.

Question 16

Working: Y=stress/strain=(4x10^7)/(2x10^-4)=2x10^11 Pa.

Final answer: 2 x 10^11 Pa.

Question 17

Working: B=-Delta p/(Delta V/V), so Delta p=-B(Delta V/V)=-(1.5x10^9)(-2x10^-4)=3x10^5 Pa.

Final answer: 3 x 10^5 Pa.

Question 18

Working: From Y=FL/(A Delta L), rearrange Delta L=FL/(AY). If F, A and Y stay fixed, Delta L is directly proportional to L; doubling L doubles the extension.

Final answer: DeltaL proportional to L.

Question 19

Working: The proportional limit is the end of the straight-line stress-strain region where stress is proportional to strain. The elastic limit is the greatest stress after which a material still returns fully to its original dimensions when unloaded; the elastic limit can be beyond the proportional limit.

Final answer: Proportional limit ends linearity; elastic limit is the greatest stress for full recovery after unloading.

Question 20

Working: Start with Y=FL/(A Delta L), giving Delta L=FL/(AY). Substitute: Delta L=50(1.5)/[(10^-6)(2x10^11)]=75/(2x10^5)=3.75x10^-4 m.

Final answer: DeltaL=FL/(AY)=3.75 x 10^-4 m.